SoftwareONE

EMPOWERING COMPANIES
TO TRANSFORM

axis DATA

AGENDA

01 Assessment details

02 Recommended path forward

03 Next steps

01

Assessment Details

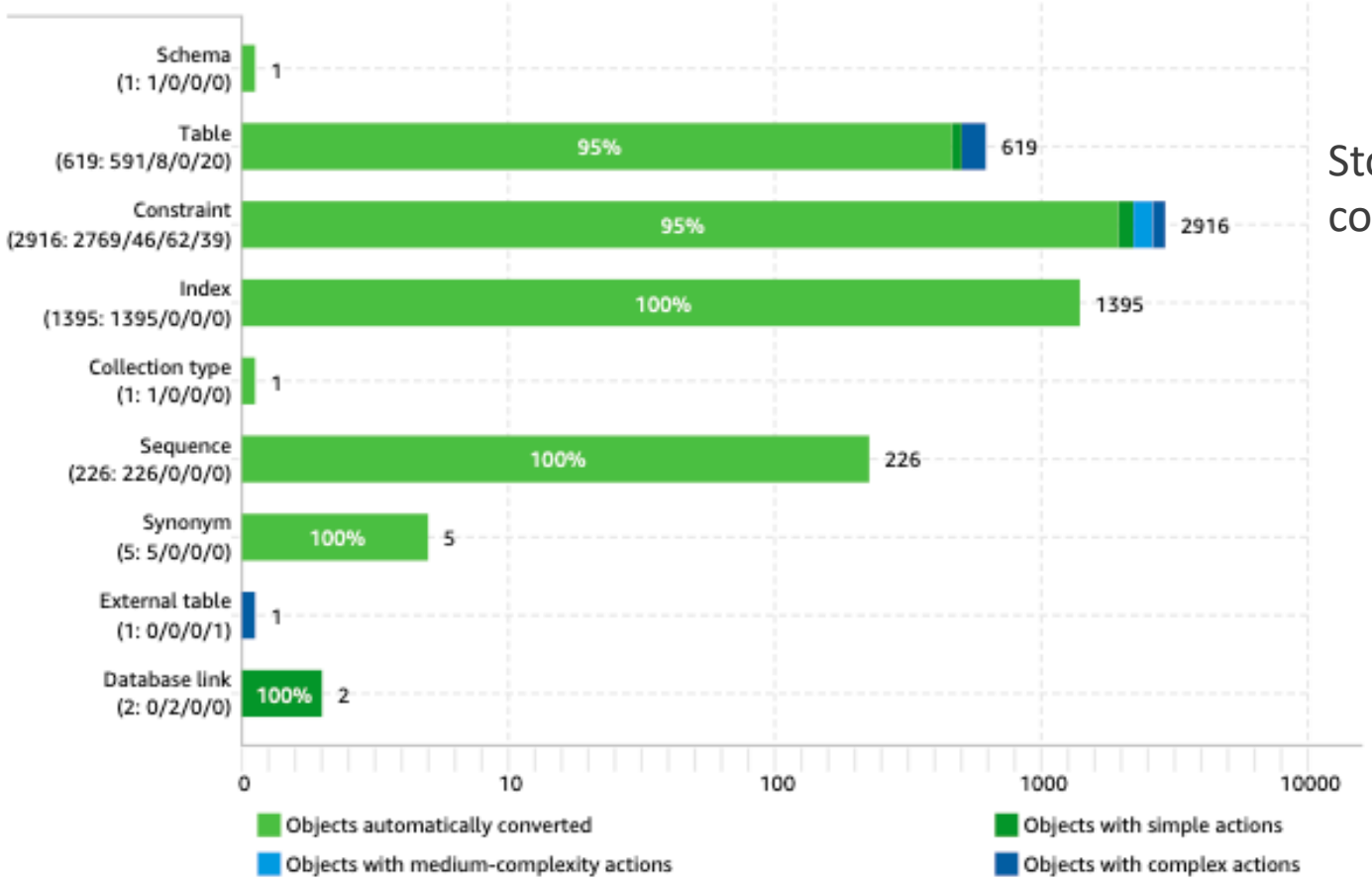
Assessment Process

Custom Queries	Schema Conversion Tool	ORA2PG
DATABASE LANDSCAPING	MIGRATION COMPLEXITY	MIGRATION EFFORT
INSTABILITY & LoC	MANUAL ACTIONS	MANUAL ACTIONS

Assessment Results

Storage Object Conversion Statistics TKMTS

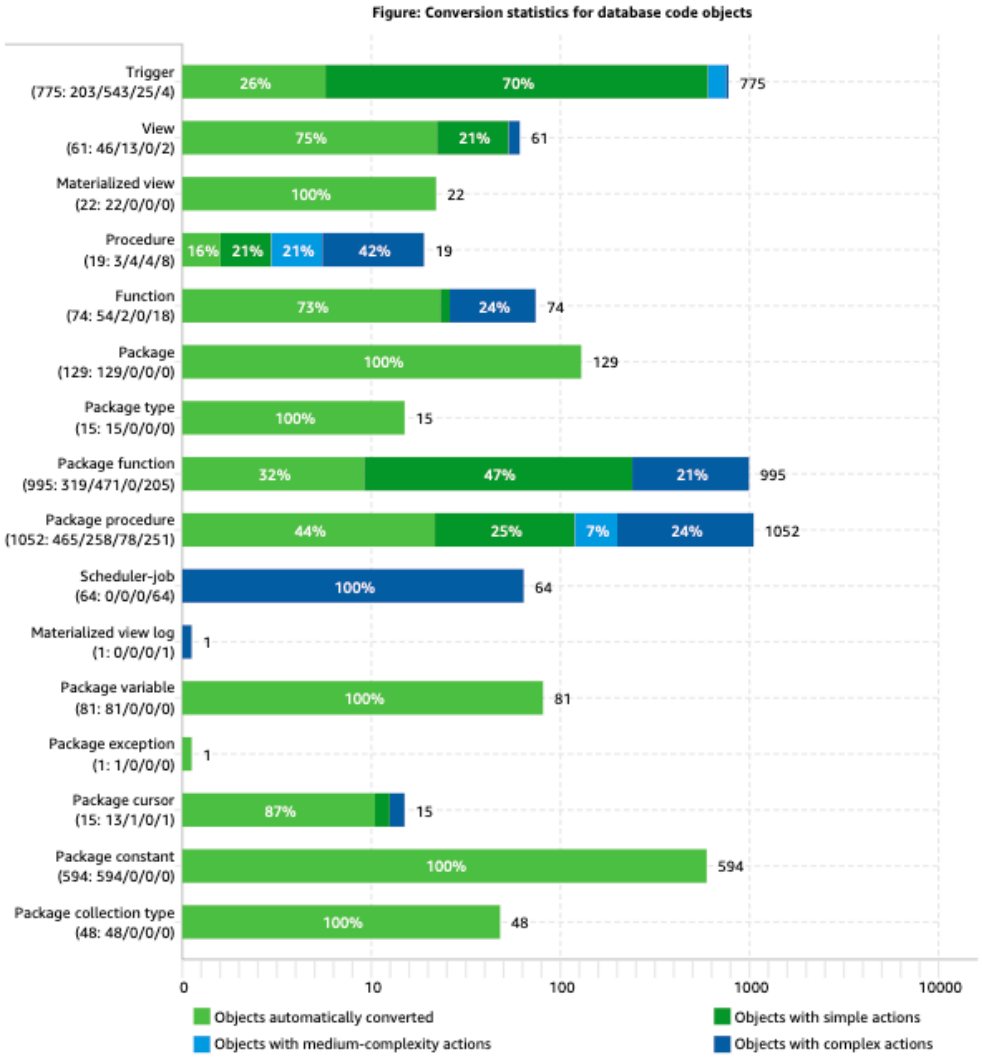
Figure: Conversion statistics for database storage objects



Storage objects require less intervention, and are concentrated in 5% of tables and 5% of constraints.

Assessment Results

Storage Object Conversion Statistics TKMTS

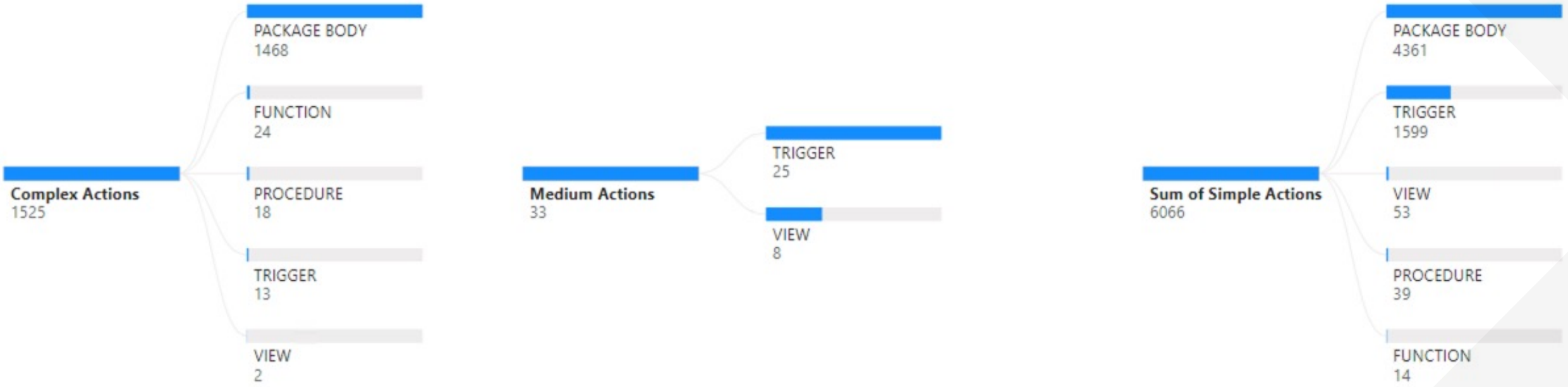


Digging deeper into the analysis of procedures, functions, triggers, and packages, it is possible to visualize that 84% of procedures must be manually intervened to execute a migration effort from Oracle to Postgres.

8 of 19 procedures require complex manual interventions. 15.7% of procedures can be converted without manual intervention.

Assessment Results

Migration Manual Activities



Assessment Results

Landscape

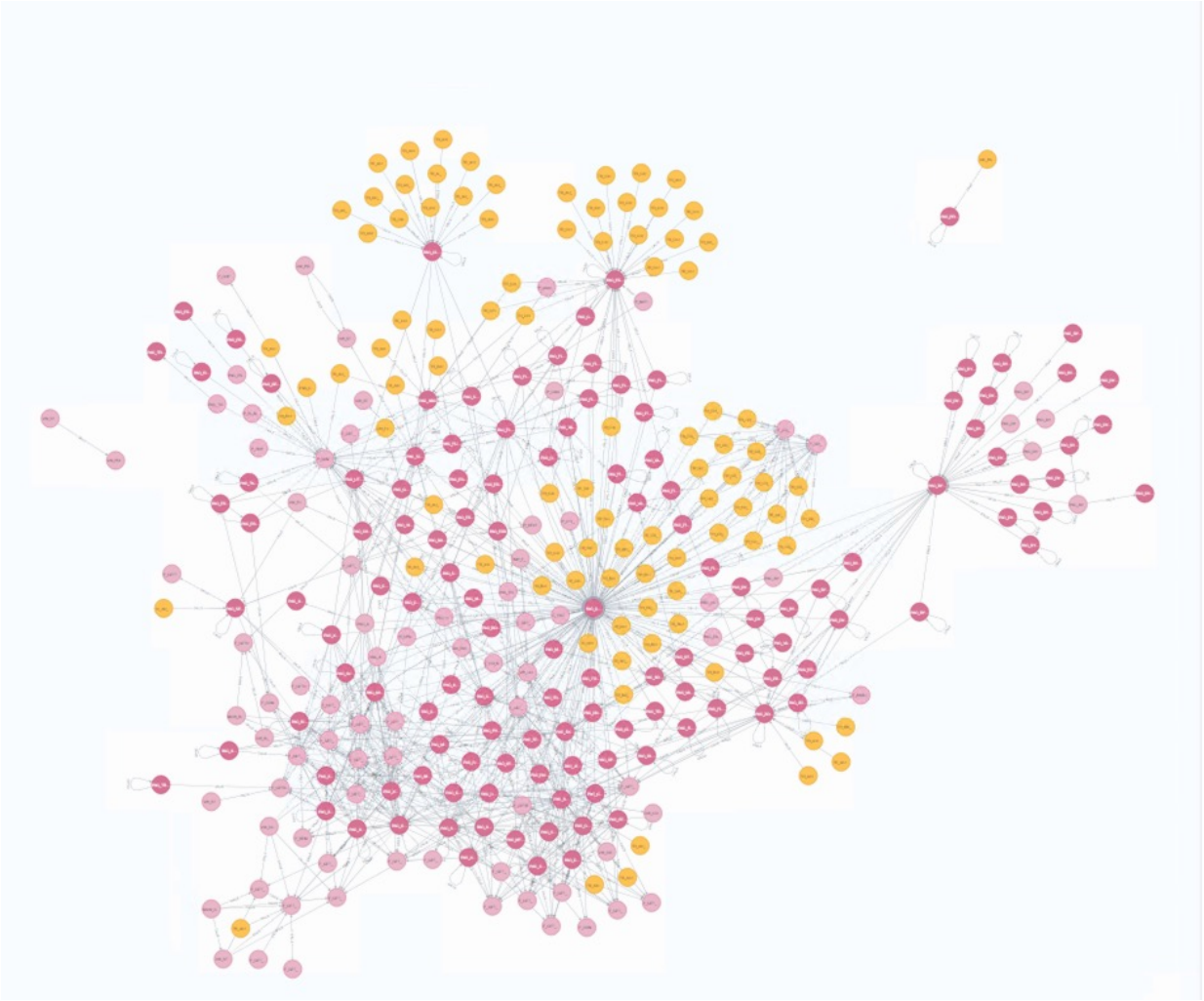
682 Tables	24 Procs.	775 Triggers	91 Functions
67 Views	141 Packages	22 M. Views	3 Java Classess

10 TB size of BD

Oracle 19c

Assessment Results

Coupling



Overview

Node labels

* (313) CodeObject (313) Package (129) Trigger (98)

Relationship types

* (946) CALLS (946)

Displaying 313 nodes, 946 relationships.

Coupling

This diagram shows the coupling between Code objects, based on the references and dependencies between them.

We have included labels to facilitate the differentiation of the objects mapped in the graph.

Assessment Results

Coupling

Overview

Node labels

* (343) CodeObject (343) Tightly (206) MediumCoupling (65) Loosely (72)

Relationship types

* (1159) CALLS (1159)

Displaying 343 nodes, 1,159 relationships.

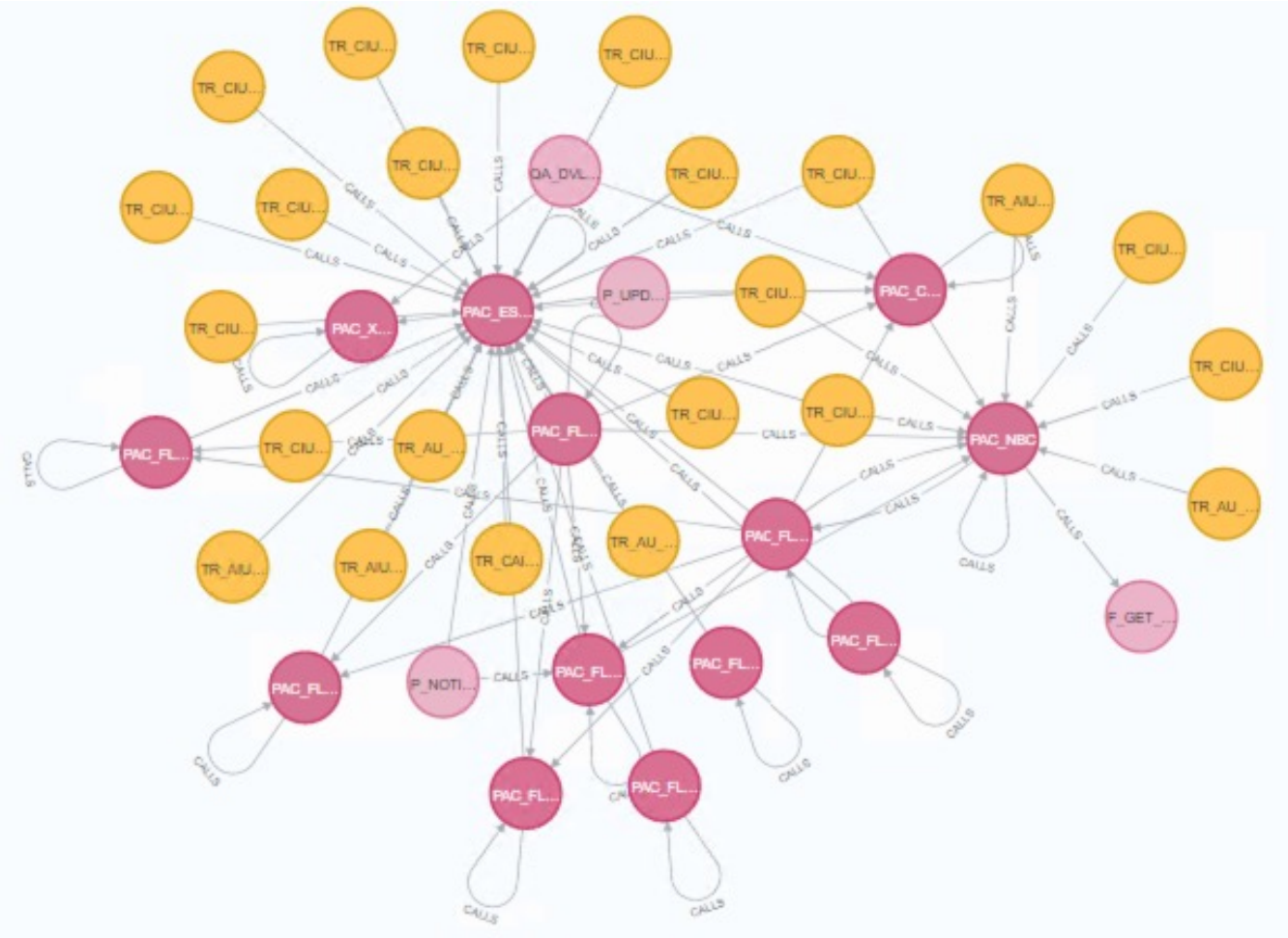
Coupling

This diagram shows the coupling between Code objects, based on the references and dependencies between them.

We have included labels to facilitate the differentiation of the objects mapped in the graph.

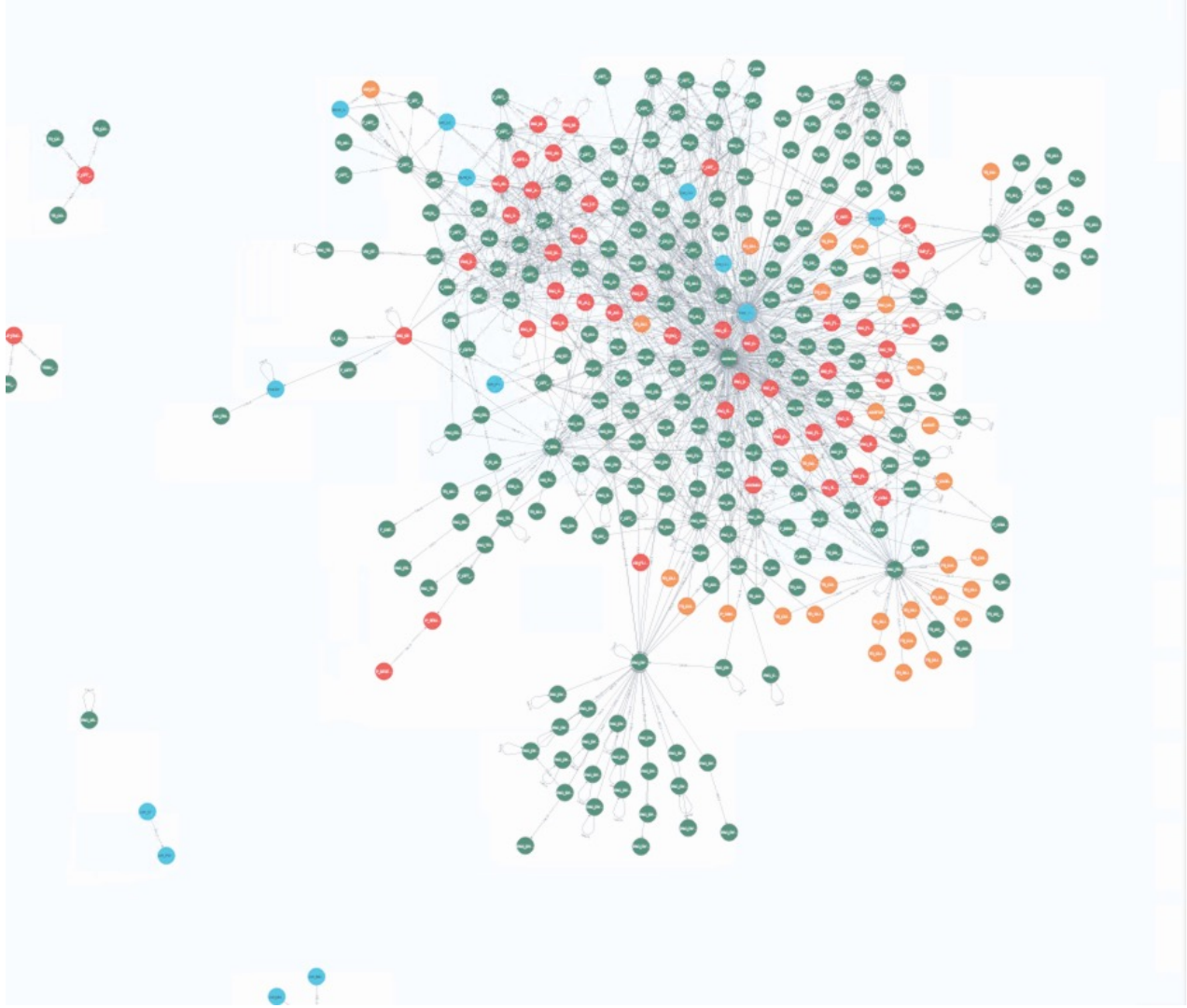
Assessment Results

Communities


TR_CIU...
TR_CIU...
TR_CIU...

Assessment Results

Complexity



Overview

Node labels

* (343) | CodeObject (343) | Complex (53) | Simple (244) | Medium (31) | Noaction (15)

Relationship types

* (1159) | CALLS (1159)

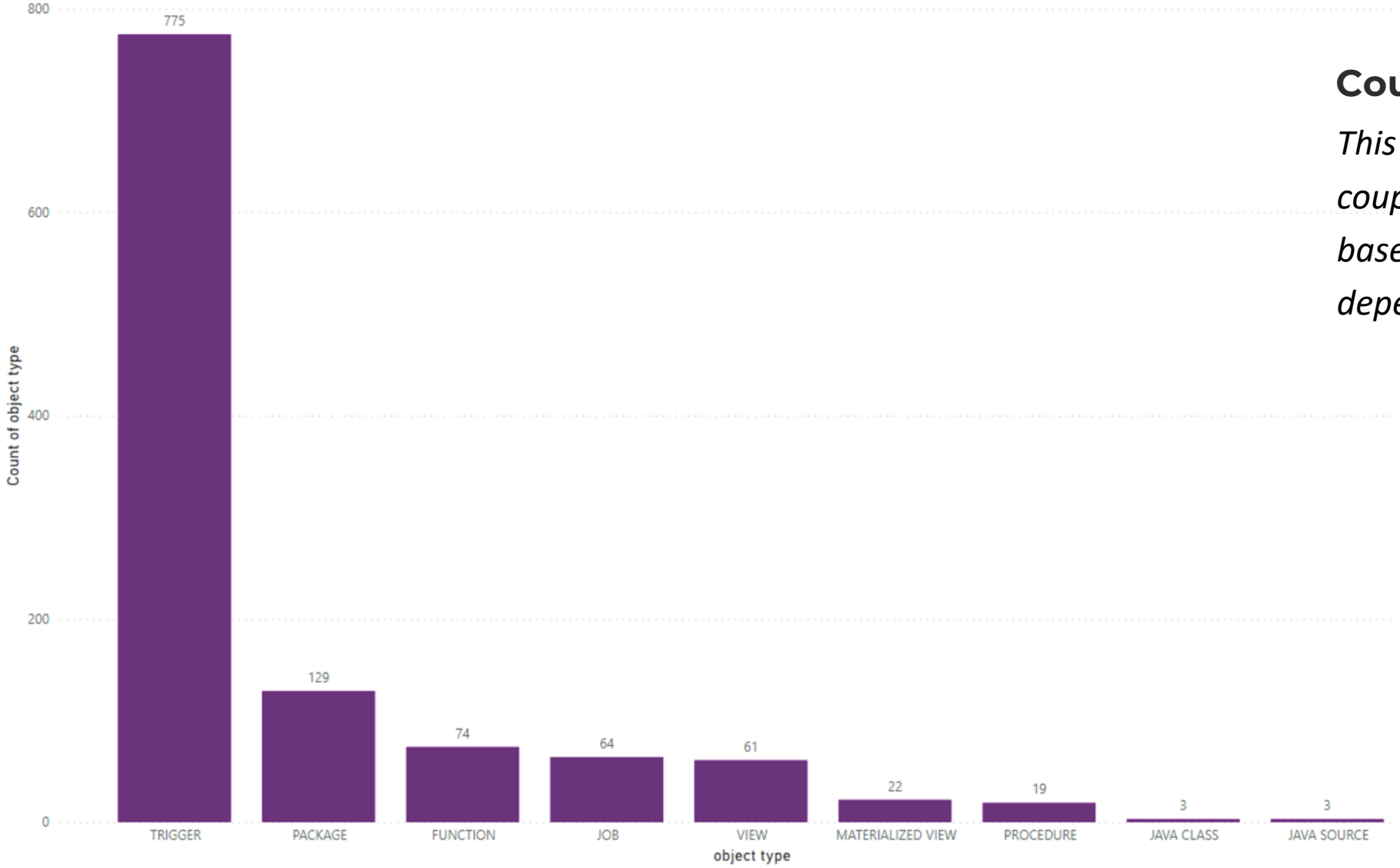
Displaying 343 nodes, 1,159 relationships.

Communities

Making use of GDS algorithms, we have done an analysis that allows us to identify groups of nodes that are highly connected to each other and have a weak connection with nodes outside that group.

Assessment Results

Counting objects by type

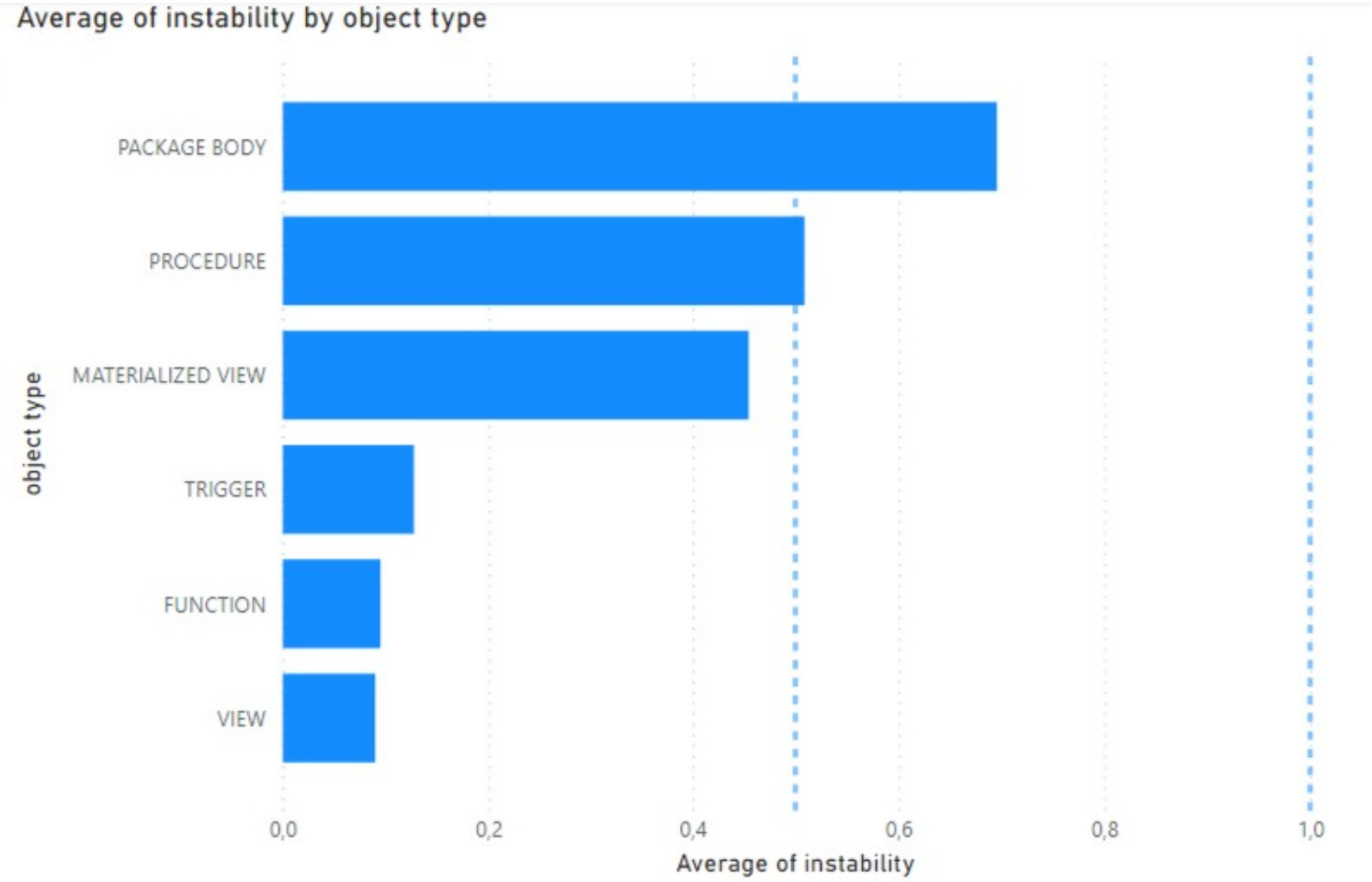


Coupling

This diagram shows the coupling between Code objects, based on the references and dependencies between them.

Assessment Results

Coupling

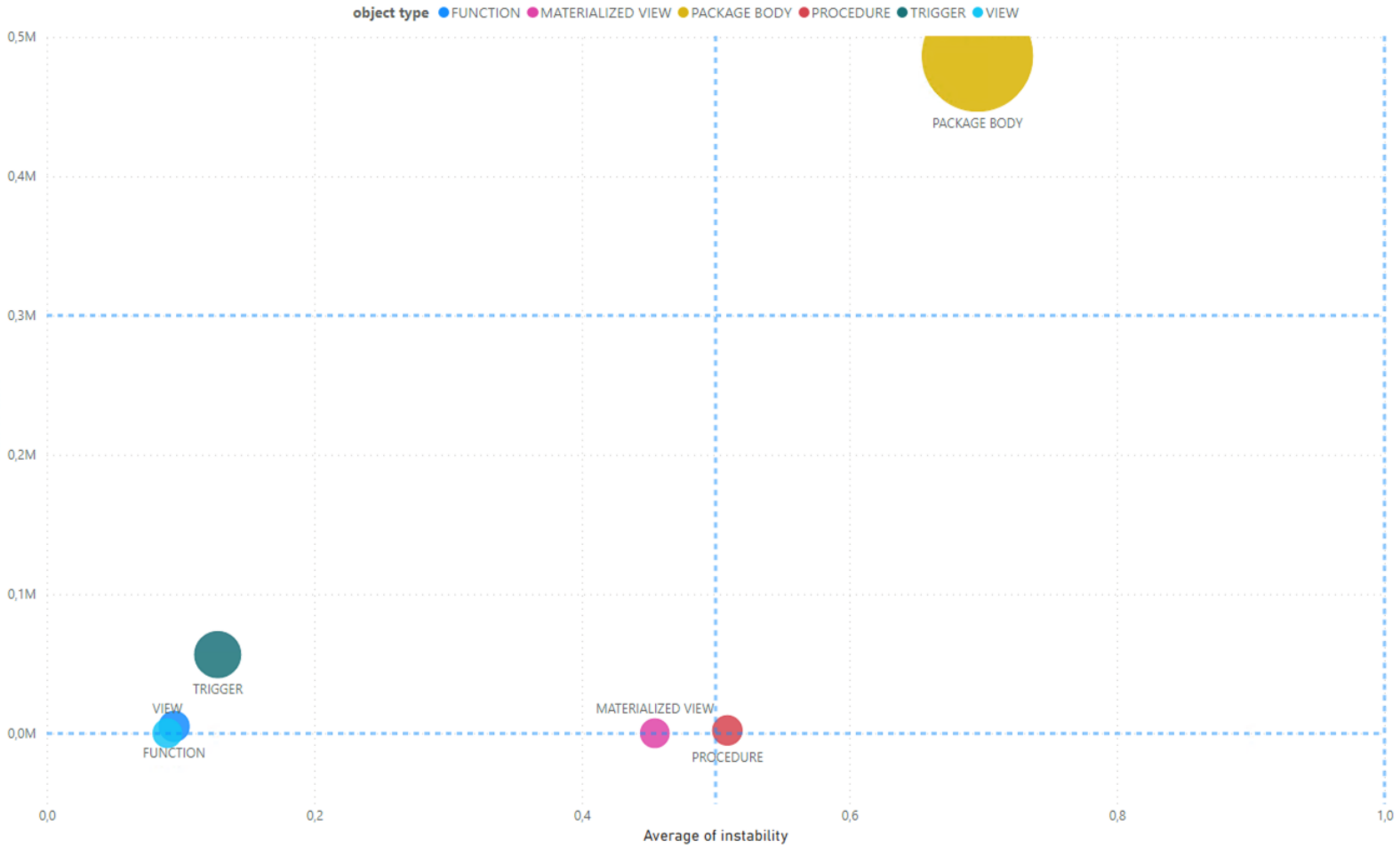


Instability vs Object Type

The components are considered to be generally stable when this calculation places the value of I closer to 0.

Assessment Results

Coupling

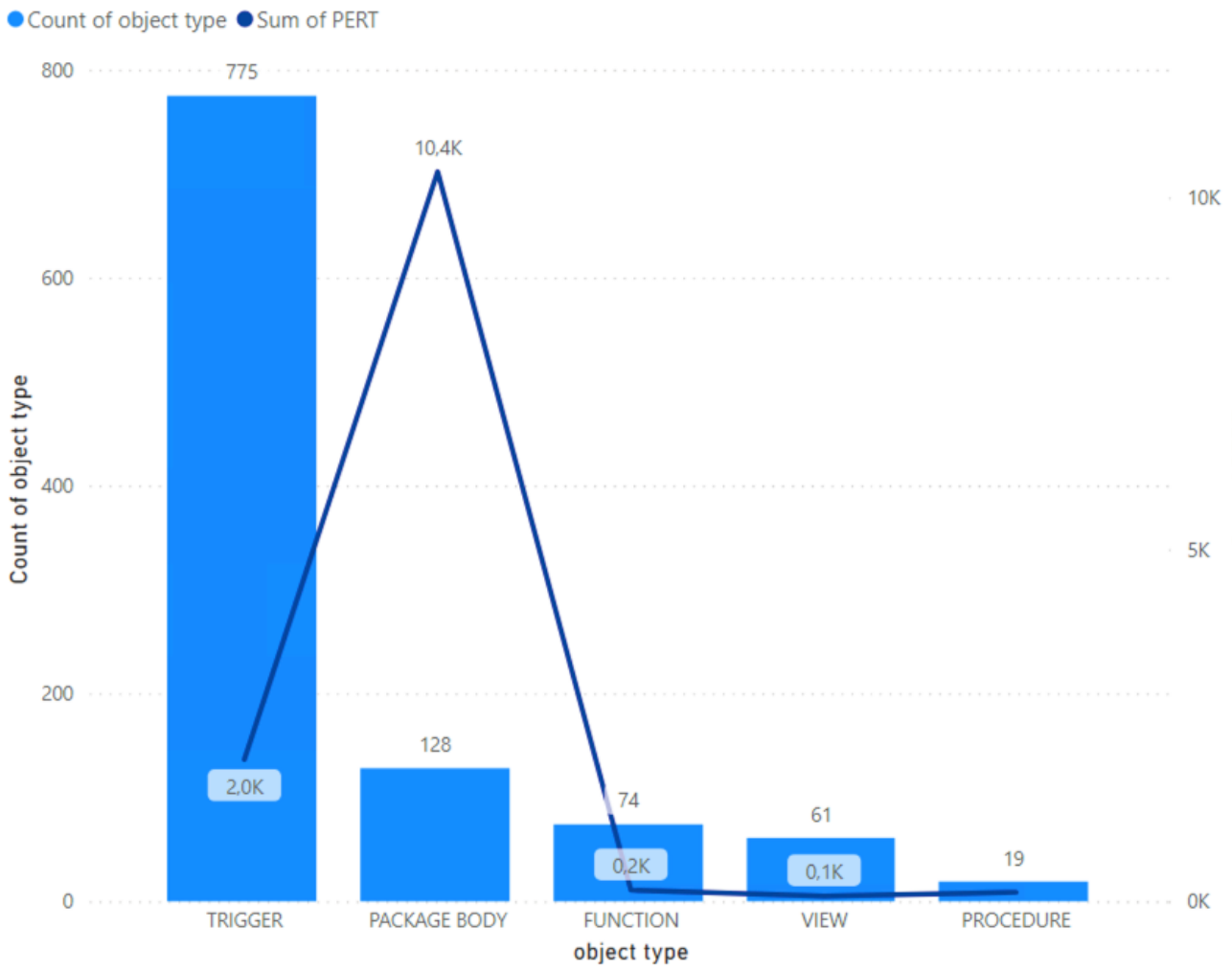


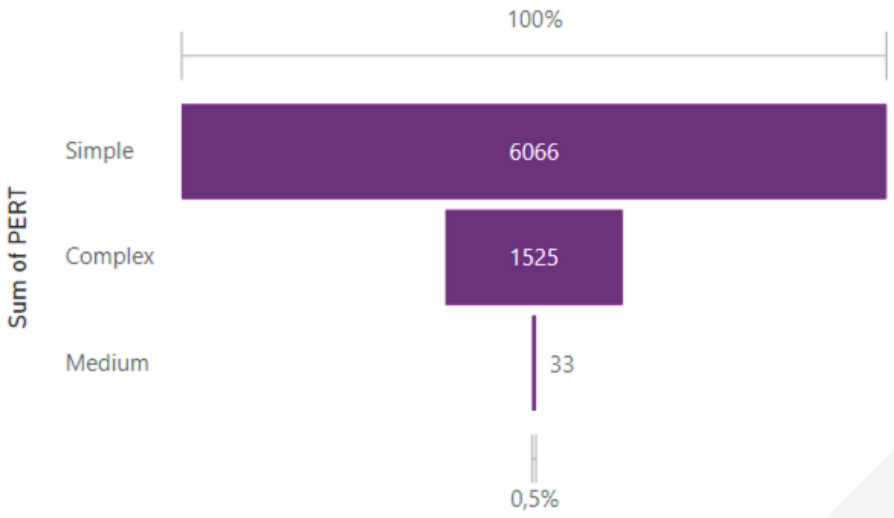
Instability vs LoC

The first quadrant shows objects that show a low level of coupling and a smaller number of lines of Code.

Assessment Results

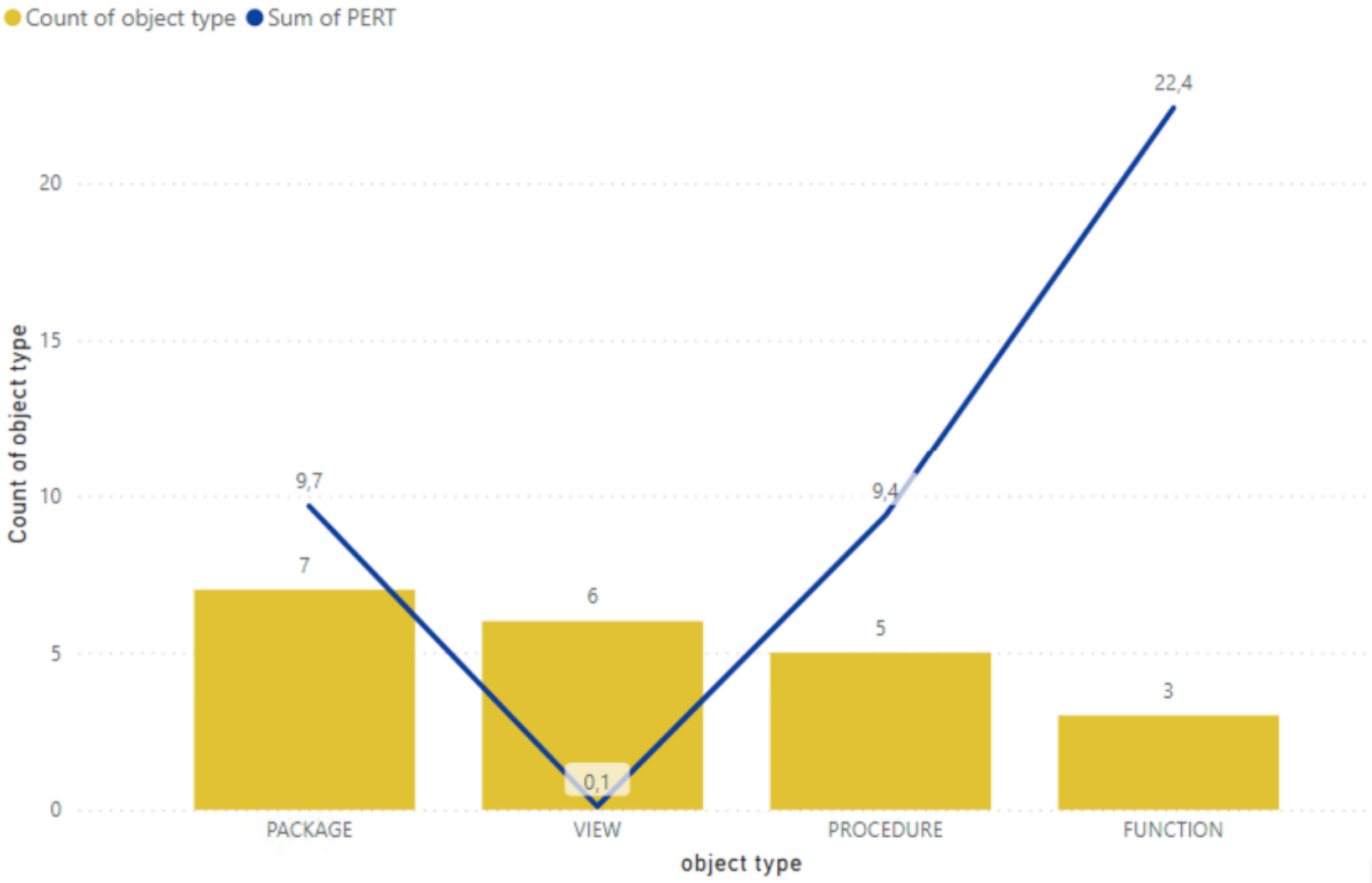
Estimated migration effort





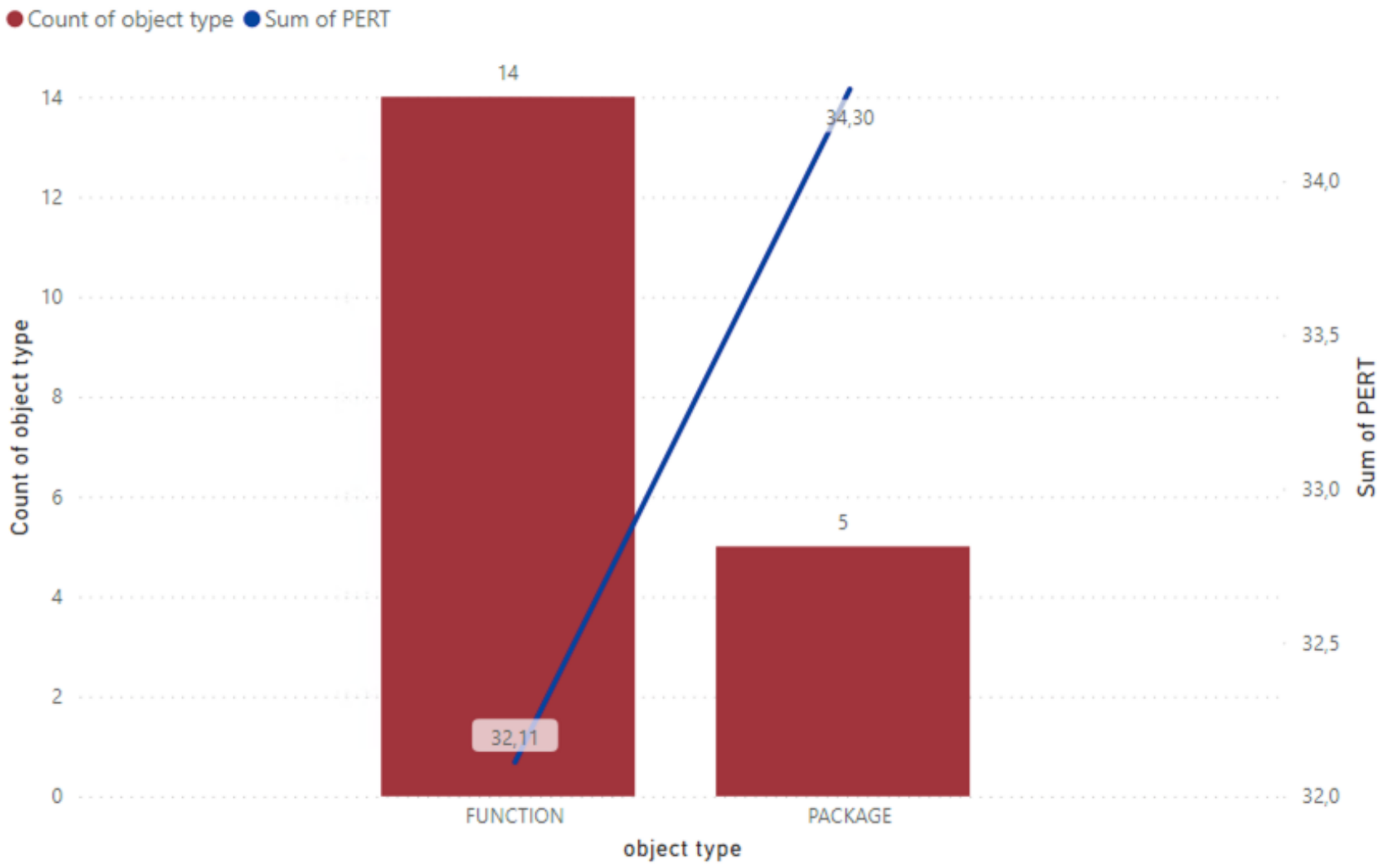
Assessment Results

Estimated migration effort



Assessment Results

Estimated migration effort



Assessment Results

Estimated migration effort

SWO Estimate (P)	SCT Estimate (M)	ORA2PG Estimate (O)
17884 Total Hours	**12248** Total Hours	**9585** Total Hours

PERT =(O+(4*M)+P)/6

PERT Estimate

12743 Total Hours

- **DRY**
- **REUSE**
- **REMOVE**

02

Recommended Path to Follow

Recommended Path to Follow

Incremental Modernization Plan

Stage 1

Approach:
Application-layer logic

Process:
DDD and/or Communities
DevOps Maturity
Implement business logic in ESB by domains and/or communities

Outcome:
Code object business logic deployed at the application tier

Stage 2

Approach:
Oracle to PostgreSQL

Process :
Migration from SO to PostgreSQL/SCT
Oracle Cutover

Outcome :
Storage objects fully migrated to Postgres

Stage 3

Approach:
Landing Zone

Process :
Deploy transitional architecture
Build/deploy LB microservices
Update CI/CD pipelines

Outcome :
Microservices by business domain in transitional architecture

Stage 4

Approach:
Architecture Modernization

Process :
Deploy event-driven architecture

Outcome :
Decoupled application supported by an event processing subsystem

Migration

Modernization

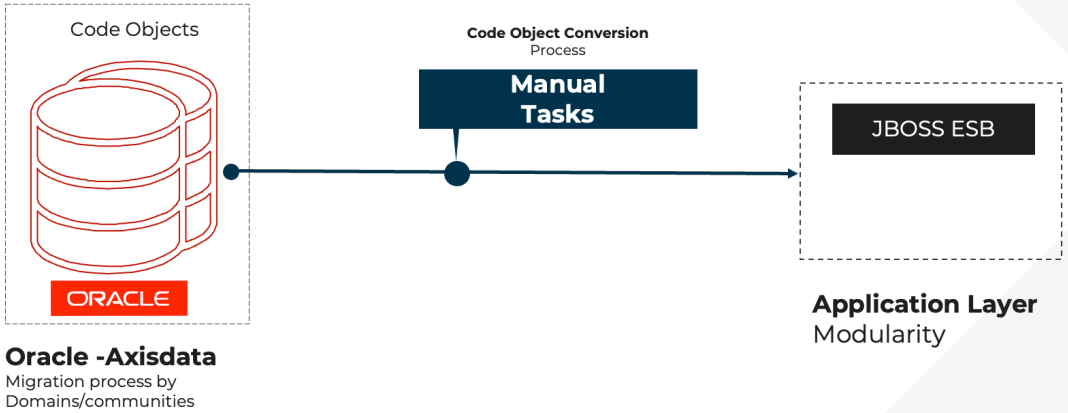
Modern Software Development Practices

Recommended Path to Follow

Stage 1

Manual Migration

- Identification of Restrictions
- DDD
- DevOps Maturity
- BL Implementation on ESB

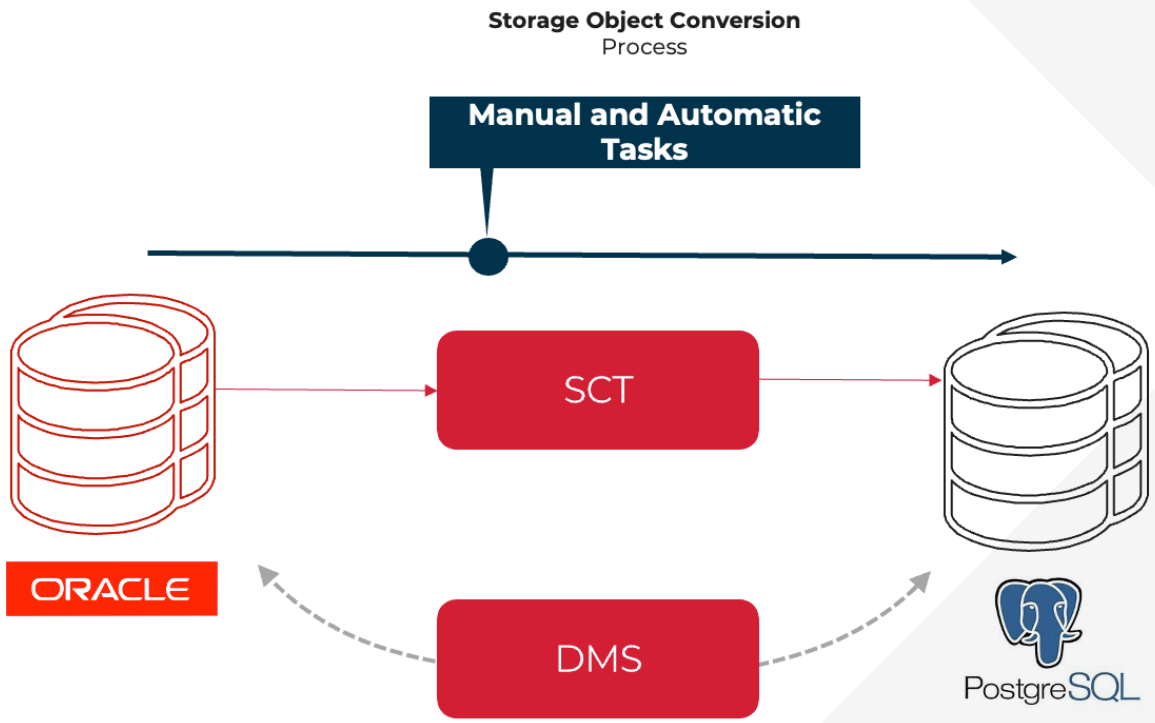


Recommended Path to Follow

Stage 2

Automatic/Manual Migration

- Identification of Restrictions
- Migration of Storage Objects.
- Cutover

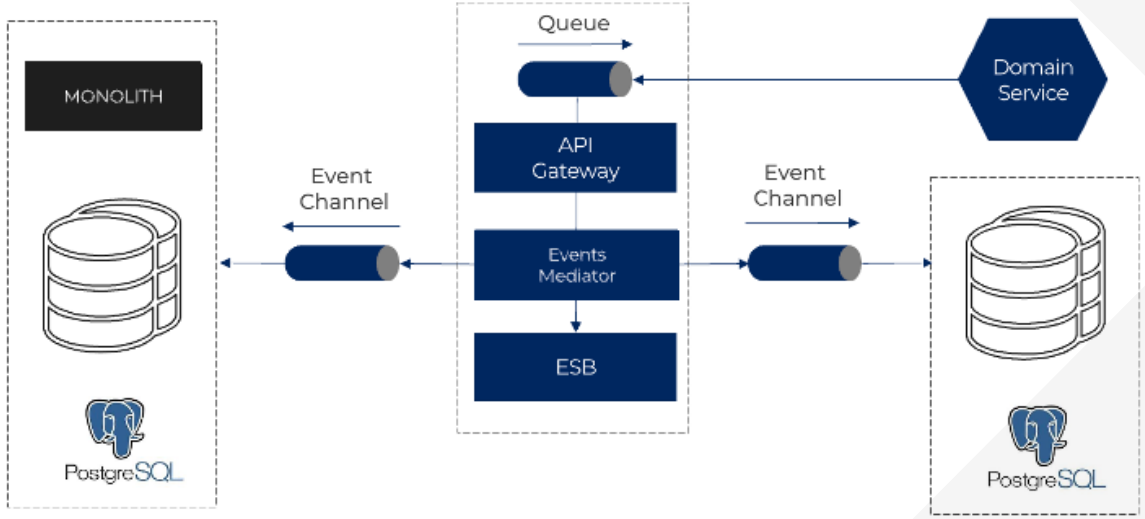


Recommended Path to Follow

Stage 3

Landing Zone

- Identification of Restrictions
- Transitional Architecture
- Microservices Deployment
- DevOps



Recommended Path to Follow

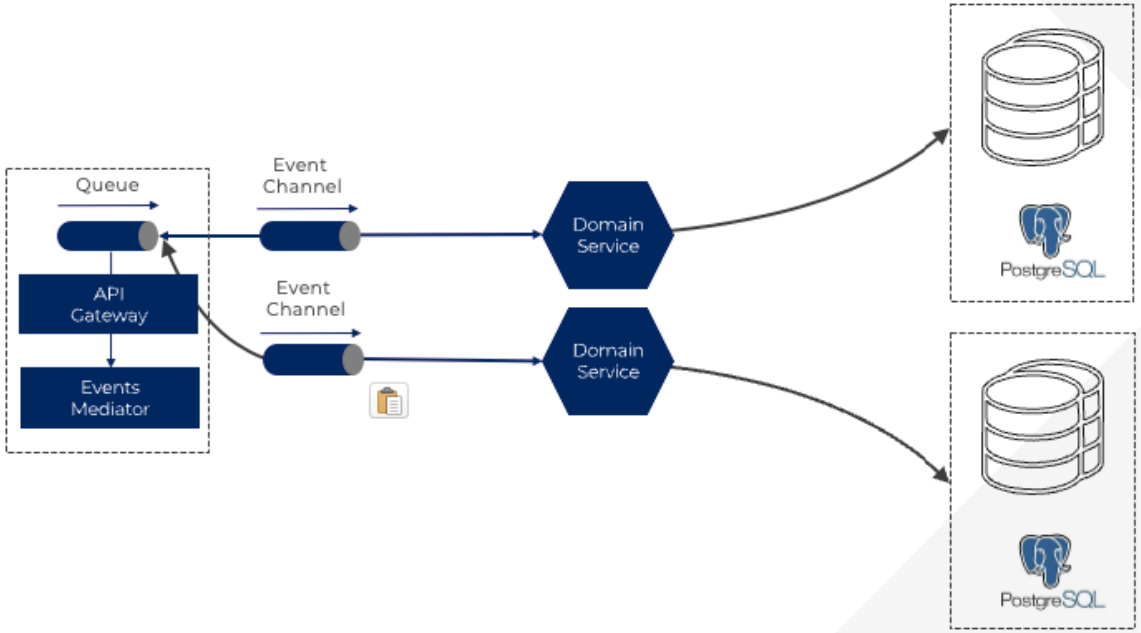
Stage 4

Architecture Modernization

Identification of Restrictions

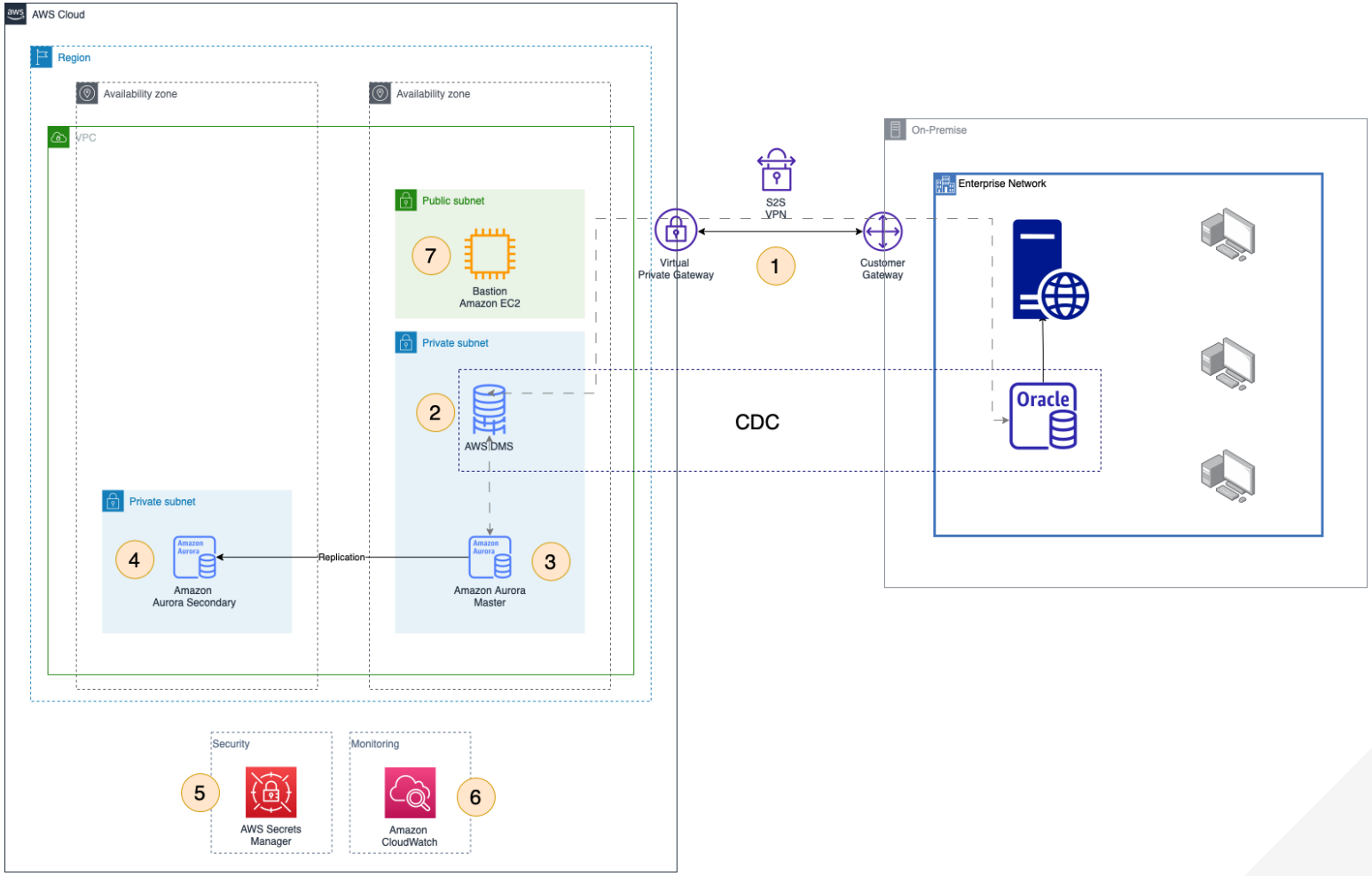
Event-Driven/MSs architecture

DevOps



Recommended Path to Follow

Reference Architecture



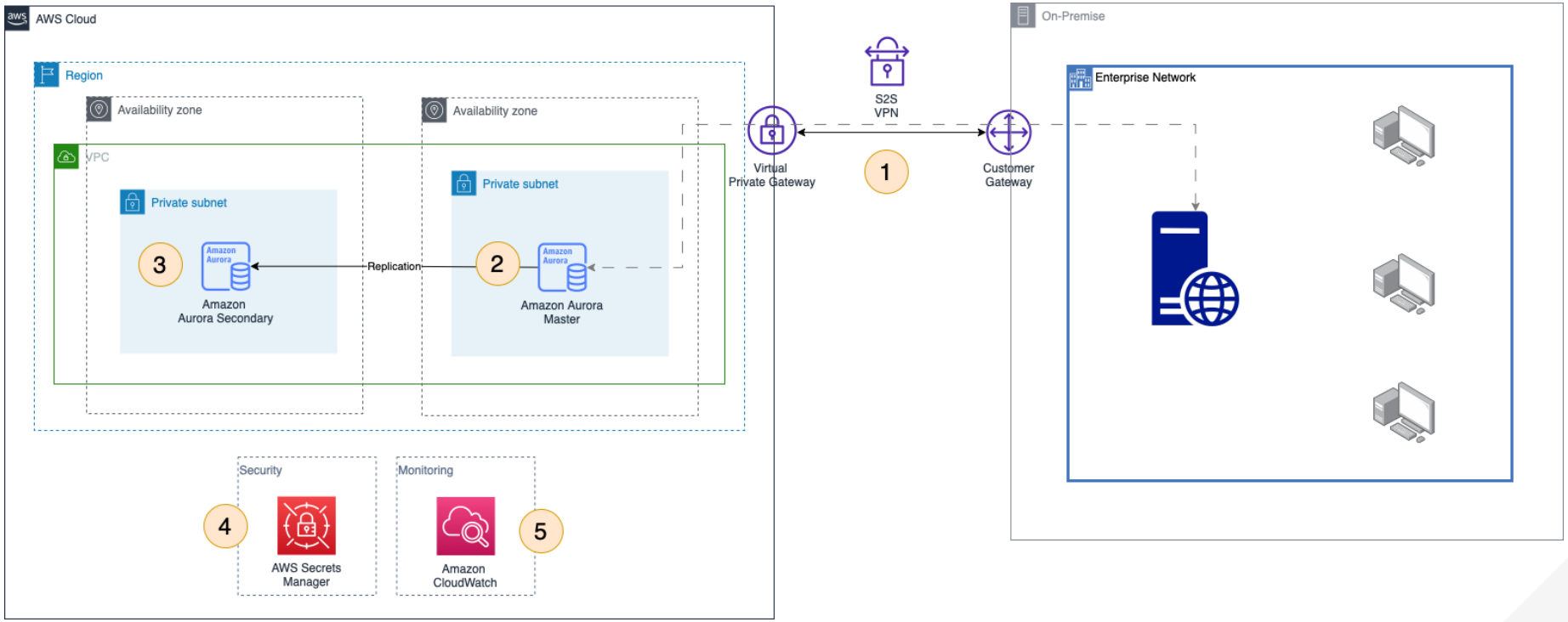
Recommended Path to Follow

Reference Architecture

Region	Description	Service	Monthly	First 12 months total	Curr.
Europe (Paris)	Axisdata Networking	VPN Connection	$ 1.319,0	$ 15.828,0	USD
Europe (Paris)	Axisdata Database	Amazon Aurora PostgreSQL-Compatible DB	$ 2.641,5	$ 31.698,2	USD
Europe (Paris)	Axisdata DMS	AWS Database Migration Service	$ 6.358,9	$ 76.306,6	USD
Europe (Paris)	Axisdata Monitoring	Amazon CloudWatch	$ 22,2	$ 266,2	USD
Europe (Paris)	Axisdata Secrets	AWS Secrets Manager	$ 5,4	$ 64,8	USD
Total			$ 10.347,0	$ 124.163,8	

Recommended Path to Follow

Reference Architecture



Recommended Path to Follow

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Europe (Paris)	Axisdata Monitoring	Amazon CloudWatch	$ 22,2	$ 266,2	USD
Europe (Paris)	Axisdata Secrets	AWS Secrets Manager	$ 5,4	$ 64,8	USD
Total			$ 3.988,10	$ 47.857,20	

03

Next Steps

Commercial and Next Steps

DDD Activity Plan

Activity	Description	Resources
Architecture Mapping	Perform an architectural analysis to document and examine the current structure of the application, with the aim of understanding the interconnection and interactions between the different components and subsystems in all their layers	1 Senior Solutions Architect 1 Technical Architect Axisdata Team
Event Storming	Event storming aims to understand business processes, identifying among other things, the business domains of the legacy application, in order to break down complex tasks into simpler and more manageable tasks.	1 Senior Solutions Architect 1 Technical Architect Axisdata Team
Metrics Analysis	Measure and analyze metrics such as centrality, density, and modularity of identified communities to gain a better understanding of the structure of relationships between sources and business domains.	1 Senior Solutions Architect 1 Technical Architect
Pattern Analysis	Search identified communities for valuable insights into business domains and to identify potential migration issues or opportunities.	1 Arquitecto de Soluciones Senior 1 Arquitecto Técnico
Definition of Waves	Prioritization of migration waves (release plan) according to the results of interpolation of business domains and communities according to identified metrics and patterns. This allows you to define the order of deployment of the business logic.	1 Senior Solutions Architect 1 Technical Architect
Migration Plan	Build the migration plan based on the waves defined according to the prioritization and restrictions identified during the previous activities	1 Senior Solutions Architect 1 Technical Architect Axisdata Team